JACOB (JAKE) BLAUSTEIN

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EDUCATION

The University of Texas at Austin — Bachelor of Science | *Mechanical Engineering* — May 2028
Minor | *Entrepreneurship*
Overall GPA 4.0/4.0 | SAT 1580

The London School of Economics — Certificate in Entrepreneurial Finance — Summer 2025

EXPERIENCE

SuperDial – *Agent Operations Intern*; New York, NY — May 2026 – Aug 2026
Healthcare AI startup developing LLM voice agents to automate provider-insurer phone calls for revenue cycle management
- Managed end-to-end operations for customers representing ~$3M ARR, owning core success, quality, and delivery metrics
- Drove 13% call success rate improvements for key customers by resolving systemic failures and designing AI call logic
- Built Python/SQL operations tooling, including an automated LLM QA system, customer dashboards and portal UI updates
- Oversaw a 400+ person team of human callers and auditors to optimize call completion, quality control, and delivery speed

Underwood Laboratory at UT Austin – *Undergraduate Researcher*; Austin, TX — Nov 2025 – Present
- Support research on reactive transport in fluids and its applications in propulsion, hypersonics and advanced thruster design
- Contribute to design, fabrication, and testing of experimental hardware for novel air-breathing very low Earth orbit thrusters
- Build a synthetic waveform generation framework in Python to train a variational autoencoder for pulse-shape optimization

New York University Langone Health – *Research Assistant*; New York, NY — Nov 2022 – June 2024
- Researched effects of historical redlining in NYC on particulate pollution and health disparity due to cardiovascular impacts
- Tested use of air purifiers to mitigate cardiovascular risk by analyzing 600,000+ lines of air quality data using R-Studio
- Published findings in 2 first-author papers in the journal Circulation Research and presented at 3 national conferences

LEADERSHIP EXPERIENCE & ACTIVITIES

Texas Marine Robotics – *Mechanical Design Team Co-Lead* — Jan 2026 – Present
- Design mechanical components for an autonomous surface vessel, supporting sensor integration and weight optimization
- Develop CAD models for mounting hardware and structural components for stability in demanding marine environments
- Conduct CFD and physical flow simulations to analyze hydrodynamic performance and inform structural design decisions

Personal Engineering Project: Thermal Search Quadcopter Drone — Oct 2025 – Present
- Design custom quadcopter drone using SolidWorks-modeled frame geometry and 3D-printed carbon fiber PLA components
- Wire and program brushless motors, ESCs, sensors, and Arduino-based flight controller for stable flight performance

Delta Tau Delta – *Philanthropy Chair* — Aug 2024 – Present
- Manage a $50K fundraising campaign to support a child and his family affected by childhood cancer via the B+ Foundation
- Organize an 8-chapter basketball tournament, coordinating event logistics and raising $22.5K for pediatric cardiac care
- Coordinate weekly volunteering at East Austin elementary schools for tutoring young students from disadvantaged areas

London School of Economics: Entrepreneurial Finance — Summer 2025
- Developed a mock startup company and full VC pitch, including detailed financial modeling and strategic growth planning
- Built comprehensive cash flow projections, capitalization tables, and return analysis to evaluate different funding scenarios
- Applied valuation methods such as the venture capital method and comparable market analysis to assess early-stage startups

HONORS AND AWARDS

- National Merit Scholar — 2024
- Outstanding Presentation Award, First Place, NYC Cardiovascular Research Conference — 2023

PUBLICATIONS

- **Blaustein JR,** Quisel M, Wittkopp S. Environmental Impacts on Cardiovascular Health & Biology. *Circ. Res*. (2024)
- **Blaustein JR**, Moon H, Newman JD, Wittkopp S. Residents of historically redlined districts in New York City experience more indoor particulate air pollution, increasing cardiovascular risk. *Circulation Res*. (2024); 135:e1-e3

SKILLS & INTERESTS

Skills: SolidWorks, Python, Matlab, SQL, RStudio, 3D Printing, Powerpoint, Word, Excel
Interests: Golf, Tennis, IM Basketball, Escape Rooms, Carpentry, Rock Climbing, Travel, Skiing, Pickleball, NYT Puzzles